

COMPENSATED GRAM CORRECTIONS FOR ZERO COUNTS OF THE RIEMANN ZETA FUNCTION

R. ARUN CHANDRU

ABSTRACT. We develop a compensated Gram-matrix correction for counting zeros of the Riemann zeta function. A single averaged matrix witness combines overlapping local windows without an outer-block loss. Its admissible local floor depends on the spectral clipping threshold; this dependence permits a stronger correction for the union of simple and critical-line zeros and for low-multiplicity counts than for the simple-critical intersection. Combining the finite argument with unconditional pair correlation and two computer-assisted seven-point inequalities gives lower limiting proportions exceeding 0.888554068993217696 for simple-or-critical zero mass, 0.837368508521536857 for mass of multiplicity at most two, and 0.939197203148497742 for locations of multiplicity at most two among distinct locations. Each improves an explicitly identified same-statistic comparison. We also obtain a simple-critical bound, a total-distinct bound, fixed higher-multiplicity-cutoff families, and qualified short-height extensions. The analytic deductions and the finite certification rules are given explicitly. Supplement information treats the numerical certificates and a local thirteen-point reduction; that reduction is not used as a whole-domain inequality or as a further zero-count improvement.

CONTENTS

1. Introduction	2
1.1. Counts and main conclusions	3
1.2. Comparison with earlier bounds	3
1.3. Relation to earlier methods	5
1.4. Organization and the role of computation	6
2. A finite trace inequality with a free clipping threshold	6
2.1. The operator attached to a conjugation-invariant multiset	6
2.2. Exceptional orbits and positive inertia	7
2.3. The excess that the different counts measure	8
3. One witness for all overlapping windows	8
3.1. The variational formula	8
3.2. The threshold-dependent cap and pressure transfer	9
3.3. A bounded state correction	10
4. Multiplicity accounting with the simple Gram block retained	11
4.1. Mass of multiplicity at most two	11
4.2. Costs per location and the distinct denominator	12
5. The two kernels and their certified local inequalities	12
5.1. Normalization and the exact energy cost	12
5.2. The local inputs	13
6. From the finite operator to zeta zeros	14
6.1. The fixed-test pair-correlation input	14
6.2. Smoothing without changing the test with height	15
7. Evaluation of the global proportions	16

Date: 10 September 2026.

2020 Mathematics Subject Classification. Primary 11M26; Secondary 15A42, 65G20.

Key words and phrases. Riemann zeta function, Gram matrix, zero multiplicity, pair correlation.

7.1. The simple-critical and total-distinct counts	16
7.2. The union and higher-multiplicity mass	16
7.3. Distinct low-multiplicity locations	16
7.4. Exact arithmetic and strict comparison signs	17
8. Short-height intervals with fixed support margin	17
9. Scope of the accompanying local extension	19
Acknowledgments	19
Note on AI Use	19
References	19

1. INTRODUCTION

The distinction between a zero and a zero location is essential in questions about multiplicity. So is the distinction between being simple *and* lying on the critical line and being simple *or* lying there. These questions share analytic information, but they spend that information in different ways. A method that improves one count need not improve another. This work tries to make that distinction effective through a finite Gram-matrix argument.

The starting point is the unconditional pair-correlation theorem of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [4]. The recent arguments of Alpöge–Furman and Lamzouri [1, 11] connect this analytic input to the simultaneous simplicity and criticality of more than two thirds of the zeros. Local Gram corrections, including computer-assisted weighted overlap inequalities, subsequently appear in the public work of Ainta and Shi [2, 17]. Waddle’s public low-multiplicity draft [20] provides a further useful comparison: its scalar multiplicity accounting can be strengthened by retaining the entire simple-real Gram block.

Our contribution is a finite transfer mechanism and its application to these different counts. The main point is not to add unrelated local estimates to a spectral bound. Instead, all overlapping windows are represented inside one admissible matrix witness. The quadratic cost of that witness pays for their overlap. A row-sum estimate then controls the windows that require clipping. Since that control grows as the square of the clipping threshold, a local floor that is too large for the simple-critical argument can still be used for the union and multiplicity arguments.

There are two independent numerical inputs. One is Ainta’s seven-point cosine-kernel inequality, which we credit as prior work. The other uses an explicit two-mode perturbation of the same density. The perturbation raises the analytic energy, so its local improvement is useful only after both effects have been included in the same calculation. We give the exact formulas, rather than combining the energy of one kernel with the local certificate of another. The computer-assisted inputs are whole-domain inequalities, not sampled minima. Their finite acceptance rules and reproducibility requirements are supplied in the accompanying supplement.

All results in this article concern ordinary nontrivial zeros of ζ . Neither the Riemann hypothesis nor a narrow-box assumption on the zeros is used in the global conclusions. The short-height statements additionally invoke the stated theorem of Wang [21], currently a preprint; the global conclusions do not depend on it.

1.1. Counts and main conclusions. Write a nontrivial zero location as $\rho = \beta + i\gamma$, and write m_ρ for its multiplicity. All sums in this subsection run over distinct locations with $0 < \gamma \leq X$. Define

$$(1.1) \quad N(X) = \sum_{\rho} m_{\rho}, \quad n(X) = \#\{\rho : m_{\rho} = 1, \beta = \tfrac{1}{2}\},$$

$$(1.2) \quad N^d(X) = \sum_{\rho} 1, \quad U(X) = \sum_{m_{\rho}=1 \text{ or } \beta=1/2} m_{\rho},$$

$$(1.3) \quad M_{\leq r}(X) = \sum_{m_{\rho} \leq r} m_{\rho}, \quad D_{\leq r}(X) = \sum_{m_{\rho} \leq r} 1.$$

Thus U is a mass count of a union, whereas n counts the intersection. The denominator for $D_{\leq r}$ below is N^d , not N .

Here are the exact constants used throughout. Put

$$(1.4) \quad z = \frac{1}{\sqrt{2}}, \quad R_0 = \frac{1}{2} + z \cot z, \quad u_1 = \frac{13}{500}, \quad u_2 = -\frac{7}{500},$$

$$(1.5) \quad \Delta = \sum_{j=1}^2 \left(\frac{1}{2} - \frac{1}{4\pi^2 j^2} \right) u_j^2, \quad R_1 = R_0 + \Delta,$$

$$(1.6) \quad a_0 = \frac{19}{5000}, \quad d_0 = \frac{1}{500}, \quad a_1 = \frac{7}{400}, \quad d_1 = \frac{3}{250}, \quad T_1 = R_1 + d_1,$$

$$(1.7) \quad \Gamma = 4 + 2\sqrt{2(1-a_1)}, \quad Z = \frac{\Gamma^2}{8} = 3 - a_1 + 2\sqrt{2(1-a_1)}, \quad \beta = T_1 - 1 - a_1, \quad E = \frac{\beta}{Z}.$$

In particular $0 < E < 1/2$. The conclusions proved below are

$$(1.8) \quad \liminf_{X \rightarrow \infty} \frac{n(X)}{N(X)} \geq C := \frac{2 - R_0 - d_0}{1 - a_0} > 0.673058325315610967,$$

$$(1.9) \quad \liminf_{X \rightarrow \infty} \frac{N^d(X)}{N(X)} \geq \frac{1+C}{2} > 0.836529162657805483,$$

$$(1.10) \quad \liminf_{X \rightarrow \infty} \frac{U(X)}{N(X)} \geq 1 - 2E > 0.888554068993217696,$$

$$(1.11) \quad \liminf_{X \rightarrow \infty} \frac{M_{\leq 2}(X)}{N(X)} \geq \frac{3 - T_1}{2 - a_1} > 0.837368508521536857,$$

$$(1.12) \quad \liminf_{X \rightarrow \infty} \frac{D_{\leq 2}(X)}{N^d(X)} \geq 1 - \frac{T_1 - 1 - a_1}{8 - 2T_1 - a_1} > 0.939197203148497742.$$

The strict decimal inequalities compare fixed real constants. They do not assert that every finite-height ratio already exceeds the printed decimal. For each fixed integer $r \geq 3$, the same argument also gives

$$(1.13) \quad \liminf_{X \rightarrow \infty} \frac{M_{\leq r}(X)}{N(X)} \geq 1 - \frac{r+1}{r-1}E,$$

$$(1.14) \quad \liminf_{X \rightarrow \infty} \frac{D_{\leq r}(X)}{N^d(X)} \geq 1 - \frac{E}{r-1-rE}.$$

The word “fixed” matters: no uniform asymptotic assertion with a height-dependent cutoff is intended.

1.2. Comparison with earlier bounds. The union, multiplicity-two mass, and multiplicity-two distinct-location conclusions improve identified unconditional bounds for the same counting statistics. The union bound improves Lamzouri’s Theorem 1.1 [11]; the two multiplicity-two bounds improve

the displayed formulas in Waddle’s public research draft [20]. The respective improvements, as differences of proportions, exceed

$$0.000934060819863356, \quad 0.001118156681831034, \quad 0.000469272388685036.$$

Multiplying these differences by 100 gives percentage-point gains. The comparisons use the exact expressions, not subtraction of rounded headline percentages.

To state the comparisons compactly, put

$$(1.15) \quad C_0 = 2 - R_0, \quad C_1 = \frac{3 - R_0}{2}, \quad B = (3 - 2\sqrt{2})(R_0 - 1), \quad C_2 = 1 - 2B.$$

Thus $C_0 = 0.672500703679411\dots$ is the Montgomery–Taylor simple-critical constant and $C_1 = 0.836250351839705\dots$ is the corresponding distinct-zero constant. Alpöge and Furman [1, Theorem A] established these bounds unconditionally by an inertia and rank–trace argument for a finite compression of Weil’s Hermitian form. Lamzouri [11] recovered them through a Hilbert-space argument and obtained the union constant

$$C_2 = \frac{1 + 2\sqrt{2} + 2C_0}{3 + 2\sqrt{2}} = 0.887620008173354\dots$$

Ratio	Present lower bound	Named comparison
U/N	$1 - 2E$ > 0.888554068993217696	Lamzouri [11]: $C_2 = 0.887620008173354\dots$
$M_{\leq 2}/N$	$\frac{3 - T_1}{2 - a_1}$ > 0.837368508521536857	Waddle [20]: $C_1 = 0.836250351839705\dots$
$D_{\leq 2}/N^d$	$1 - \frac{\beta}{8 - 2T_1 - a_1}$ > 0.939197203148497742	Waddle [20]: $1 - \frac{R_0 - 1}{2(4 - R_0)}$ $= 0.938727930759812\dots$
n/N	$\frac{2 - R_0 - d_0}{1 - a_0}$ > 0.673058325315610967	Devine [8]: $673399/10^6 = 0.673399$ (stated unconditional theorem)
N^d/N	$\frac{3 - R_0 - d_0 - a_0}{2(1 - a_0)}$ > 0.836529162657805483	Alpöge–Furman and Lamzouri [1, 11]: $C_1 = 0.836250351839705\dots$

TABLE 1. Five separate global lower limiting proportions. Each exact expression in the middle column bounds the corresponding lower limit from below; the strict inequalities compare these expressions with shorter decimals. All external statements listed here are unconditional. Waddle’s source is an unrefereed public draft and Devine’s is a public preprint. The last row is a comparison with the stated foundational bounds, not a claim of an overall distinct-zero record.

Here n counts simple critical zeros, whereas U counts with multiplicity the union of simplicity and critical-line location. The denominators N and N^d are different: $M_{\leq r}/N$ is a fraction of zero mass, and $D_{\leq r}/N^d$ is a fraction of distinct complex zero locations. Locations with the same imaginary part are not identified. These conventions also match the external statements in Table 1. A dyadic statement for additive counts gives the corresponding cumulative lower limit by summing sufficiently high dyadic pieces of $(0, X]$; the bounded initial remainder disappears after division by the same denominator.

For every fixed integer $r \geq 3$, Waddle’s displayed mass and distinct-location bounds are respectively

$$1 - \frac{(r+1)B}{r-1}, \quad 1 - \frac{B}{r-1-rB}.$$

Our corresponding formulas replace B by E , and the proved inequality $0 < E < B < 1/2$ makes both improvements strict. The mass gain is $(r+1)(B-E)/(r-1)$; the location comparison follows from the strict increase of $x/(r-1-rx)$ on the relevant interval. This conclusion uses the retained multiplicity budget. It is not a formal consequence of a union bound: a multiset supported on real locations of multiplicity $r+1$ has full union mass and no mass at multiplicity at most r .

The fixed-cutoff statement is an improvement over a specified family. It does not assert the best available bound for every cutoff. Simonič, Trudgian and Turnage-Butterbaugh [18, Theorems 4 and 7] give unconditional exponential estimates for the number of locations of exact multiplicity j . Weighting by j and summing the tail gives mass bounds tending to one as the cutoff grows, whereas the displayed mass family here tends to $1 - E < 1$. Consequently the exponential-tail bound is stronger for sufficiently large fixed cutoffs.

The simple-critical conclusion refines Ainta’s public bound [2], whose exact expression is

$$\frac{1345000C_0 - 2680}{1340003} = 0.673008527927779 \dots$$

It remains below Shi’s stated research-draft candidate $0.6733169771424713 \dots$ [17] and Devine’s stated 0.673399 [8]. The latter preprint’s numerical proof has not been independently reconstructed here. These are external comparison targets, not premises of our proof, and we claim no overall simple-critical or distinct-zero record. Zeta Lab’s public report [12] also records an eight-point value $0.67305298298962888 \dots$, distinguishing an interval-certified local inequality from its Lean implication conditional on that inequality. The public union extension [13] supplies another explicit treatment of the distinction between union and intersection.

Stronger numbers with additional hypotheses must be compared separately. The union bounds $8/9$ in [5, Remark 2] and $(2 + C_0)/3 = 0.890833567893 \dots$ discussed in [11, Remark 1.2] use a narrow-box assumption on the horizontal locations of all sufficiently high dyadic zeros; they are not unconditional union comparators. Under RH, Gonçalves, de Laat and Leijenhorst [9, Theorem 1] prove $\liminf M_{\leq 2}/N \geq 0.9614$. Their cutoff-three value 0.9787 also carries the theorem’s additional qualification concerning Maple summation of specified series. RH-dependent pair-correlation estimates in [7] include 0.6792 for simple zeros and 0.8477 for distinct zeros. Devine’s claims beyond 0.6792 likewise require additional density or horizontal-profile assumptions [8]. None of these numbers is an unconditional obstruction to the bounds proved here.

1.3. Relation to earlier methods. The analytic framework and several finite ingredients have established antecedents. Montgomery’s pair-correlation theorem [14] and the Montgomery–Taylor window reported in [15] underlie the constant R_0 ; the associated extremal problem is treated in [6]. Aryan [3] obtained an unconditional form of the relevant Fejér-kernel calculation, and Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [4] proved the published full-height pair-correlation formula used here. Lamzouri’s finite-multiset formulation and removal of the rational pair weight [11] provide the direct analytic interface. Wang [21] supplies the separate short-height input, with support strictly smaller than the fixed interval exponent. We prove no new pair-correlation theorem and do not insert a height-dependent test into a fixed-test error estimate.

Ainta [2] introduced the clipped spectral stability correction and quantitative overlapping seven-point windows used in this line of refinements. The Fourier perturbations of the cosine density and nonuniform pair weights with distance-capacity constraints in the inspected Ojha and Shi drafts [16, 17] are also prior work. Shi’s threshold-one trace–energy envelope, pressure rescaling, and two-certificate supporting-plane deduction are directly relevant antecedents. In particular, neither weighted overlap nor removal of the elementary threshold-one cap is claimed as a new idea here. The

arbitrary-parameter scalar multiplicity framework and its uncorrected low-multiplicity consequences are credited to the two-trace development and Waddle's displayed deductions [1, 20].

The finite contribution here is the use of one common averaged witness to transfer a whole-domain local Gram inequality directly to the clipped spectral defect while preserving its pressure floor. The uniform coefficient estimate depends on the clipping threshold. For the seven-point weights used here, its admissible cap is $40t^2/2401$. This dependence is material: the new local floor $a_1 = 7/400$ is unavailable through this cap at $t = 1$, but is admissible at $t = 2$ and at the union threshold $t = 1 + \sqrt{2(1 - a_1)}$. Thus the stronger local certificate is matched to the count being proved. Retaining the resulting simple-Gram correction in the separate per-location multiplicity costs then produces the mass and distinct-location bounds from that same certificate. The elementary variational identity for clipping is part of the proof; the gain comes from its common-witness application and the uniform capacity estimate.

The required two-mode profile, its normalization and energy cost, and the whole-domain seven-point inequality are established explicitly. The local interval engine is an attributed adaptation of Ainta's MIT-licensed implementation, using Arb ball arithmetic [10]. The separate cosine row for the simple-critical and total-distinct conclusions is Ainta's completed input. Our reproduced executions and exact scalar calculations certify the stated finite numerical premises; they do not replace the finite operator argument or the passage to zero limits.

Finally, bounded-potential versions of the common-witness argument give additional finite inequalities, but the general telescoping subaction idea already occurs in the public literature, including [8]. Partial larger-window certificates are identified by their domains. They supply no global zero-count conclusion unless the entire required configuration space has been covered.

1.4. Organization and the role of computation. We first prove the finite trace and inertia inequality, then construct the common local witness. A separate multiplicity calculation retains the simple Gram block while grouping all other locations by multiplicity. Only after those arguments are complete do we introduce the two kernels and invoke pair correlation. This order exposes the interface between finite linear algebra, local numerical certification and analytic number theory.

The numerical part has two logically separate jobs. Rigorous interval enclosures certify the local inequalities on the entire nonnegative gap orthant; exact rational interval arithmetic then encloses the constants in (1.8)–(1.14). The latter calculation does not replace the former. Conversely, a local certificate has no zero-count meaning until its matching kernel has passed through the analytic limit. The supplement explains the full acceptance mathematics, the finite coverage rules and the dependence on rigorous elementary-function arithmetic.

2. A FINITE TRACE INEQUALITY WITH A FREE CLIPPING THRESHOLD

2.1. The operator attached to a conjugation-invariant multiset. Let η be a real even square-integrable function of compact support, with $\int_{\mathbb{R}} \eta(s)^2 ds = 1$. Put $f = \eta^2$ and

$$(2.1) \quad K(z) = \int_{\mathbb{R}} f(s) e^{-2\pi i z s} ds, \quad v_z(s) = \eta(s) e^{-2\pi i z s}.$$

Compact support makes K entire. It is even, real on the real axis, and $K(0) = 1$. We take the inner product to be linear in its first variable, so that

$$\langle v_z, v_w \rangle = K(z - \bar{w}).$$

The rank-one notation $v \otimes w^*$ denotes the operator $h \mapsto \langle h, w \rangle v$.

Let $\mathcal{Z}$ be a finite conjugation-invariant multiset in $\mathbb{C}$, with total multiplicity N . A location and each of its repeated copies are distinguished when forming the following sum:

$$(2.2) \quad A = \sum_{z \in \mathcal{Z}} v_z \otimes v_z^*.$$

Conjugation invariance gives $A = A^*$, and every summand has trace one. All calculations may be made in the finite-dimensional span of the vectors that occur, without any assertion of their linear independence. In particular,

$$(2.3) \quad \text{Tr } A = N, \quad H := \|A\|_{\text{HS}}^2 = \sum_{z, w \in \mathcal{Z}} K(z - w)^2.$$

For the second identity, if $A_z = v_z \otimes v_z^*$, then $\text{Tr}(A_z A_w) = K(w - z)K(z - w) = K(z - w)^2$. Summation and selfadjointness give the identity. The square in this formula is a complex square, not the absolute square of each summand. Individual summands need not be nonnegative; the total is a squared Hilbert–Schmidt norm.

Let $x_1 \leq \dots \leq x_n$ be the simple real locations in $\mathcal{Z}$. Define

$$(2.4) \quad P = \sum_{j=1}^n v_{x_j} \otimes v_{x_j}^*, \quad M = (K(x_i - x_j))_{1 \leq i, j \leq n}, \quad Q = A - P.$$

The Gram matrix M is positive semidefinite and has diagonal one. Its order is n , even if its rank is smaller. The nonzero spectra of M and P agree, and $\text{Tr } P = n$.

2.2. Exceptional orbits and positive inertia. Let r_R count the multiple real locations, of respective multiplicities $k_j \geq 2$, and let r_C count the unordered nonreal conjugate pairs, of respective multiplicities $\ell_j \geq 1$ at each member of a pair. Set $b = r_R + r_C$. Each real multiple contributes one positive rank-one term to Q . For a nonreal pair write

$$g_z = (v_z + v_{\bar{z}})/2, \quad h_z = (v_z - v_{\bar{z}})/(2i).$$

Its contribution is

$$\ell_z(v_z \otimes v_z^* + v_{\bar{z}} \otimes v_{\bar{z}}^*) = 2\ell_z(g_z \otimes g_z^* - h_z \otimes h_z^*).$$

Consequently $Q = Q_a - Q_b$ with $Q_a, Q_b \geq 0$ and $\text{rank } Q_a \leq b$. On $\ker Q_a$ the quadratic form of Q is nonpositive, so the number of its positive eigenvalues satisfies

$$(2.5) \quad n_+(Q) \leq b.$$

These nonorthogonal orbit terms are used only to prove this inertia bound. The square completion below instead uses the spectral positive and negative parts of Q , which are orthogonal.

For $t \geq 0$ define the clipped Gram defect

$$(2.6) \quad D_t(M) = \text{Tr}((M - I)^2 - (M - (1 + t)I)_+^2).$$

Here the positive part is defined spectrally. Expanding the first square and retaining the zero eigenvalues of M gives the exact identity

$$(2.7) \quad D_t(M) = \text{Tr } P^2 - n - \text{Tr}(P - (1 + t)I)_+^2.$$

The value is nonnegative: for an eigenvalue $x \geq 0$ of M , the scalar function $(x - 1)^2 - (x - 1 - t)_+^2$ is nonnegative.

For every $\Gamma \geq 2$, with $t = \Gamma/2 - 1$, we now obtain

$$(2.8) \quad H \geq \Gamma N - (\Gamma - 1)n - \frac{\Gamma^2}{4}b + D_t(M).$$

Indeed, write $Q = Q_+ - Q_-$ for its spectral decomposition. Subtracting $\Gamma \text{Tr } A$ from $\text{Tr } A^2$ gives

$$H - \Gamma N = \text{Tr } P^2 - \Gamma n + \text{Tr}(Q_+^2 - \Gamma Q_+) + \text{Tr}(Q_-^2 + \Gamma Q_- - 2PQ_-) + 2 \text{Tr}(PQ_+).$$

The last term is nonnegative. The first quadratic expression in Q_+ is at least $-\Gamma^2 b/4$, by (2.5) and scalar square completion. For a selfadjoint X and $B \geq 0$,

$$(2.9) \quad \text{Tr}(B^2 - 2XB) \geq -\text{Tr } X_+^2.$$

To see this, diagonalize X : off-diagonal entries of B add a nonnegative amount to $\text{Tr } B^2$, and each nonnegative diagonal entry minimizes its scalar quadratic at the positive part of the corresponding eigenvalue of X . Apply this with $X = P - \Gamma I/2$. Equation (2.7) completes the proof of (2.8).

2.3. The excess that the different counts measure. Write

$$(2.10) \quad u = N - n, \quad e = u - 2b = \sum_{j=1}^{r_R} (k_j - 2) + 2 \sum_{j=1}^{r_C} (\ell_j - 1) \geq 0.$$

Use the definitions of $U, N^d, M_{\leq r}, D_{\leq r}$ from the introduction, with “critical” replaced for now by “real.” The mass excluded from the union consists precisely of the nonreal pairs with $\ell_j \geq 2$. Therefore

$$(2.11) \quad N - U = 2 \sum_{\ell_j \geq 2} \ell_j \leq 2e.$$

For $r \geq 2$, writing $L_{>r} = N - M_{\leq r}$, we similarly have

$$(2.12) \quad L_{>r} \leq \frac{r+1}{r-1} e.$$

For a real location of multiplicity $k \geq r+1$, this follows from $k/(k-2) \leq (r+1)/(r-1)$. A nonreal pair of multiplicity $\ell \geq r+1$ contributes 2ℓ to the mass and $2(\ell-1)$ to e , with an even smaller ratio. Summing proves the assertion.

The distinct count has a separate exact identity:

$$(2.13) \quad N - N^d = e + r_R, \quad N^d \geq \frac{N + n - e}{2}.$$

The second assertion uses $r_R \leq b = (N - n - e)/2$. In particular it is not an inference from a bound for n alone: the excess term must be retained until the trace inequality is substituted.

Suppose, for some $a, d, B \geq 0$, a local argument has established

$$(2.14) \quad D_t(M) \geq an - dS - B, \quad S = x_n - x_1 \quad (n \geq 2), \quad S = 0 \quad (n \leq 1).$$

Substituting $b = (u - e)/2$ into (2.8) yields

$$(2.15) \quad H \geq (1+a)N - dS - B + \left(\Gamma - 1 - \frac{\Gamma^2}{8} - a \right) u + \frac{\Gamma^2}{8} e.$$

At $\Gamma = 4$ this is equivalently

$$(2.16) \quad (1-a)n \geq 2N - H - dS - B + 2e.$$

For the union, it is useful instead to choose

$$(2.17) \quad \Gamma = 4 + \sqrt{8(1-a)}, \quad 0 \leq a < 1,$$

which makes the coefficient of u exactly zero. This algebraic choice, and not a numerical sign test near zero, will be used in the limit.

3. ONE WITNESS FOR ALL OVERLAPPING WINDOWS

3.1. The variational formula. Let $G \geq 0$ be an $n \times n$ Hermitian matrix with diagonal one, and write $Y = G - I$. For $t \geq 1$,

$$(3.1) \quad D_t(G) = \max_{T=T^*, \|T\|_{\text{op}} \leq t} \{2 \text{Tr}(YT) - \text{Tr}(T^2)\}.$$

After diagonalizing Y , deleting the off-diagonal entries of T does not change the linear term, reduces its squared norm, and preserves the constraint on its diagonal. The scalar maximum is $y^2 - (y-t)_+^2$ because $y \geq -1 \geq -t$. This proves the formula. The restriction $t \geq 1$ is essential here: a symmetric norm ball would also clip some negative eigenvalues if $t < 1$.

Fix $m \geq 2$ and nonnegative symmetric coefficients c_{ij} , with zero diagonal, such that

$$(3.2) \quad \sum_{i=1}^{m-h} c_{i,i+h} \leq 2 \quad (1 \leq h < m), \quad C_{\max} := \max_i \sum_{j \neq i} c_{ij} > 0.$$

For each full consecutive block B of length m , use local indices $1, \dots, m$ and define

$$(3.3) \quad V_B = \sum_{i < j} c_{ij} |G_{B_i B_j}|^2, \quad (Z_B)_{ij} = \frac{m}{2} c_{ij} G_{B_i B_j} \quad (i \neq j), \quad (Z_B)_{ii} = 0,$$

$$(3.4) \quad s_B = \max_i \sum_j |(Z_B)_{ij}|, \quad \nu_B = \min(1, t/s_B), \quad T_B = \nu_B Z_B,$$

where $\nu_B = 1$ when $s_B = 0$. The row-sum bound gives $\|T_B\|_{\text{op}} \leq t$.

There are m partitions of the integer line into consecutive m -blocks, one for each residue of the first index modulo m . Restrict them to $\{1, \dots, n\}$, placing T_B on every full block and zero on each incomplete endpoint block. The resulting m block-diagonal matrices all have norm at most t . Their average $\bar{T}$ is therefore admissible in (3.1). Every full consecutive window occurs exactly once in this average.

The linear term is

$$2 \text{Tr}(Y \bar{T}) = 2 \sum_B \nu_B V_B.$$

For a fixed global pair of indices of separation h , its local coefficients among the windows containing it form a subcollection of $c_{1,1+h}, \dots, c_{m-h,m}$. Their sum is at most two. Weighted Cauchy–Schwarz consequently gives

$$\left(\sum_{B \ni i,j} c_{ij}^B \nu_B \right)^2 \leq 2 \sum_{B \ni i,j} c_{ij}^B \nu_B^2.$$

Since the corresponding entry of $\bar{T}$ is $\frac{1}{2} G_{ij} \sum_{B \ni i,j} c_{ij}^B \nu_B$, summing its absolute square over both orientations of each pair yields

$$\text{Tr} \bar{T}^2 \leq \sum_B \nu_B^2 V_B.$$

Thus one admissible witness proves the simultaneous estimate

$$(3.5) \quad D_t(G) \geq \sum_B J_t(B), \quad J_t(B) := (2\nu_B - \nu_B^2) V_B.$$

This is where the overlap is paid for. No sum of separate spectral pinchings, and no subsequent outer-block envelope, is required.

3.2. The threshold-dependent cap and pressure transfer. The row sum in (3.4) is controlled by the very energy that is being retained. For each row, weighted Cauchy–Schwarz gives

$$s_B^2 \leq \frac{m^2}{4} C_{\max} V_B.$$

Put

$$(3.6) \quad b_m = \frac{4}{m^2 C_{\max}}.$$

If $s_B \leq t$, then $J_t(B) = V_B$. If $s_B \geq t$, then

$$(3.7) \quad J_t(B) = \left(\frac{2t}{s_B} - \frac{t^2}{s_B^2} \right) V_B \geq b_m (2ts_B - t^2) \geq b_m t^2.$$

Now take $G_{ij} = K(x_i - x_j)$ for real ordered points, and let $g_j = x_{j+1} - x_j \geq 0$. Suppose the local kernel inequality

$$(3.8) \quad \sum_{i < j} c_{ij} |K(g_i + \cdots + g_{j-1})|^2 + \sum_{j=1}^{m-1} p_j g_j \geq \epsilon \quad (g_j \geq 0)$$

holds with $p_j \geq 0$ and

$$(3.9) \quad 0 \leq \epsilon \leq b_m t^2.$$

On an unclipped block, (3.8) bounds J_t directly; on a clipped block, (3.7) and nonnegative pressure do so. Sum over all $n - m + 1$ full windows when $n \geq m$. Each elementary gap is charged by at most one copy of each p_j , so its total coefficient is at most $P_* := \sum_{j=1}^{m-1} p_j$. Equation (3.5) then proves

$$(3.10) \quad D_t(G) \geq \epsilon n - P_* S - (m - 1)\epsilon.$$

For $n < m$, the right side is nonpositive and the assertion follows from $D_t \geq 0$. This also treats empty and one-point Gram matrices. The endpoint term is retained until the eventual height limit.

Only the unclipped domain $s_B \leq t$ needs (3.8); a whole-orthant certificate is a convenient stronger premise. The threshold condition nevertheless cannot be omitted. For the uniform choice

$$(3.11) \quad c_{i,i+h} = \frac{2}{m-h}, \quad C_{\max} = 2 \sum_{h=1}^{m-1} \frac{1}{h},$$

the endpoint row realizes the maximum. At $m = 7$,

$$(3.12) \quad b_7 = \frac{40}{2401}.$$

Our cosine floor a_0 is smaller than b_7 , whereas the two-mode floor a_1 is larger. The latter is therefore not transferred at $t = 1$ by this argument. It is transferred at $t = 2$, and at the union threshold $t = 1 + \sqrt{2(1 - a_1)} > 2$.

3.3. A bounded state correction. The common witness also permits a bounded correction that telescopes between neighboring windows. This observation is useful without asserting that an advantageous correction has been found for any particular unresolved numerical problem.

Put $q = m - 1$, and let $\Phi : \mathbb{R}_{\geq 0}^{q-1} \rightarrow \mathbb{R}$ have finite oscillation $\Omega = \sup \Phi - \inf \Phi$. For $q = 1$ its domain is a singleton. Keep $t \geq 1$, $p_j \geq 0$ and $\epsilon \geq 0$. Suppose that for every nonnegative q -tuple,

$$(3.13) \quad J_t(g) + \sum_{j=1}^q p_j g_j \geq \epsilon + \Phi(g_1, \dots, g_{q-1}) - \Phi(g_2, \dots, g_q).$$

The state differences telescope once across the complete sequence of windows. Consequently

$$(3.14) \quad D_t(G) \geq \epsilon n - P_* S - q\epsilon - \Omega.$$

For $n < m$ the right side is nonpositive and $D_t \geq 0$ again suffices. No regularity of Φ is needed. If the corresponding inequality with V in place of J_t is known only for $s \leq t$, the sufficient condition $\epsilon + \Omega \leq b_m t^2$ supplies the complementary domain using (3.7).

There is also a useful uniform stability bound. Suppose two normalized nonnegative densities have Fourier transforms differing by at most ρ on the real line. Both transforms have absolute value at most one, so (3.2) gives $|V - V'| \leq 4q\rho$ and $|s - s'| \leq mC_{\max}\rho/2$. The function $s \mapsto \min(1, t/s)$, continuously extended at zero, is $1/t$ -Lipschitz. Since $V' \leq 2q$, it follows that

$$(3.15) \quad |J_t - J'_t| \leq \left(4q + \frac{2qmC_{\max}}{t}\right) \rho.$$

Thus a proved joined inequality can be smoothed with a controlled floor loss. One must smooth that full inequality, not assume that the boundary $s = t$ remains fixed. A dilation by $\sigma > 0$ replaces Φ by $\Phi(\sigma \cdot)$ and leaves its oscillation unchanged.

4. MULTIPLICITY ACCOUNTING WITH THE SIMPLE GRAM BLOCK RETAINED

Let x_m count real locations of multiplicity m and let y_ℓ count unordered nonreal pairs of multiplicity ℓ at each member. Then $n = x_1$, $N = \sum_m m x_m + 2 \sum_\ell \ell y_\ell$, and $N^d = \sum_m x_m + 2 \sum_\ell y_\ell$. For $c \geq 1$ define

$$(4.1) \quad h_c(x) = c^2 - (c - x)_+^2 \quad (x \geq 0).$$

This is nondecreasing and concave, and $h_c(0) = 0$.

Include *all* real locations in the positive operator $P_R = \sum_{z \in \mathcal{Z} \cap \mathbb{R}} v_z \otimes v_z^*$, with multiplicity. The positive inertia of $A - P_R$ is at most $y := \sum_\ell y_\ell$. Repeating the square completion in Section 2, now with $\Gamma = 2c$, gives

$$(4.2) \quad H \geq 2cN - \text{Tr } h_c(P_R) - c^2 y.$$

For clarity, the contribution of an eigenvalue x of P_R is $x^2 - 2cx - (x - c)_+^2 = -h_c(x)$, which proves the displayed form.

Use one weighted Gram column $\sqrt{m} v_z$ per distinct real location of multiplicity m . Pinch this Gram matrix into the entire simple-real block M and one scalar block $[m]$ per multiple-real location. Scalar concavity gives

$$(4.3) \quad \text{Tr } h_c(P_R) \leq \text{Tr } h_c(M) + \sum_{m \geq 2} h_c(m) x_m.$$

Indeed, diagonalize each compressed block and use its eigenvectors as an orthonormal basis of the whole Gram space. Jensen's inequality $h_c(\langle Gv, v \rangle) \geq \langle h_c(G)v, v \rangle$ follows from the spectral measure of G at v . Summing proves (4.3). Operator concavity is not required. Gram zero eigenvalues cause no discrepancy because $h_c(0) = 0$.

Since $\text{Tr } M = n$, the exact simple-block identity is

$$\text{Tr } h_c(M) = (2c - 1)n - D_{c-1}(M).$$

Combining this with (4.2) and (4.3) produces the corrected multiplicity inequality

$$(4.4) \quad H \geq 2cN - \sum_{m \geq 1} h_c(m) x_m - c^2 \sum_{\ell \geq 1} y_\ell + D_{c-1}(M).$$

The scalar master without the final Gram term is the comparison framework in [20]. The retained term is indispensable for the improvements made here.

4.1. Mass of multiplicity at most two. Take $c = 3$, and suppose $D_2(M) \geq an - dS - B$ with $0 \leq a \leq 1/2$. In (4.4), the real-location debits are $5 - a$ for $m = 1$, 8 for $m = 2$, and 9 for $m \geq 3$; the debit per nonreal pair is 9 . Per unit mass of multiplicity at most two, these are at most $5 - a$: the four possible values are $5 - a, 4, 9/2, 9/4$. For all higher multiplicities the debit per unit mass is at most 3 . Hence

$$H + dS + B \geq 6N - (5 - a)M_{\leq 2} - 3(N - M_{\leq 2}),$$

and therefore

$$(4.5) \quad M_{\leq 2} \geq \frac{3N - H - dS - B}{2 - a}.$$

This is a mass estimate; it is not yet a proportion among distinct locations.

4.2. Costs per location and the distinct denominator. Assume a finite upper budget

$$(4.6) \quad H + dS + B \leq \mathcal{T}N, \quad \alpha = 2c - \mathcal{T}, \quad \beta = \mathcal{T} - 1 - a.$$

With $D_{c-1}(M) \geq an - dS - B$, equation (4.4) implies

$$(4.7) \quad \sum_{m \geq 1} \lambda_m x_m + \sum_{\ell \geq 1} \omega_\ell y_\ell \leq 0, \quad \lambda_m = \alpha m - h_c(m) + a \mathbf{1}_{m=1}, \quad \omega_\ell = 2\alpha\ell - c^2.$$

Suppose that for a fixed cutoff r and some $\gamma_r > 0$, these costs are at least $-\beta$ per low-multiplicity location and at least $\gamma_r - \beta$ per high-multiplicity location. A nonreal pair contains two locations, so its required bounds are twice the corresponding quantities. Then (4.7) gives $0 \geq -\beta N^d + \gamma_r(N^d - D_{\leq r})$, whence

$$(4.8) \quad \frac{D_{\leq r}}{N^d} \geq 1 - \frac{\beta}{\gamma_r}.$$

This deduction requires $N^d > 0$, as will hold at sufficiently large heights.

For $r = 2$ and $c = 3$, sufficient conditions are

$$(4.9) \quad \beta \geq 0, \quad \alpha = 6 - \mathcal{T} > 0, \quad 3 - \mathcal{T} - a \geq 0, \quad 1 - 2a \geq 0, \quad \gamma_2 = 8 - 2\mathcal{T} - a > 0.$$

Here $\lambda_1 = -\beta$, $\lambda_2 + \beta = 3 - \mathcal{T} - a$, and for $m \geq 3$

$$\lambda_m + \beta - \gamma_2 = \alpha(m - 3) \geq 0.$$

For pairs, $\omega_1 + 2\beta = 1 - 2a \geq 0$, and increasing ℓ only increases the cost. At the first high multiplicity, $\omega_3 + 2\beta - 2\gamma_2 = 9$, which remains positive for $\ell \geq 3$. These facts check every multiplicity, not a finite selection of them, and give

$$(4.10) \quad \frac{D_{\leq 2}}{N^d} \geq 1 - \frac{\mathcal{T} - 1 - a}{8 - 2\mathcal{T} - a}.$$

For $r \geq 3$, let $3 < c < 4$, and assume

$$(4.11) \quad \alpha > 0, \quad \beta \geq 0, \quad 3 - \mathcal{T} - a \geq 0, \quad 8 - 2\mathcal{T} - a \geq 0,$$

$$(4.12) \quad 4\alpha + \beta - c^2 \geq 0, \quad q_c := 2c - 1 - c^2/2 - a \geq 0, \quad \gamma_r := (r + 1)\alpha + \beta - c^2 > 0.$$

For $m = 1, 2, 3$, the values of $\lambda_m + \beta$ are respectively 0, $3 - \mathcal{T} - a$, and $8 - 2\mathcal{T} - a$. For $m \geq 4$, $\lambda_m + \beta = \alpha m + \beta - c^2$, which is nonnegative at $m = 4$ and increasing. At $m = r + 1$ it equals γ_r . For a nonreal pair the smallest value of $\omega_\ell + 2\beta$ is $2q_c \geq 0$. At $\ell = r + 1$ it exceeds $2\gamma_r$ by exactly c^2 . Thus (4.8) holds for every fixed $r \geq 3$ under the displayed conditions.

At $2c = \Gamma$ from (2.17), write $Z = \Gamma^2/8$ and $E = \beta/Z$. The identity $\Gamma - 1 - Z - a = 0$ gives $\alpha = Z - \beta$ and

$$(4.13) \quad \gamma_r = Z(r - 1 - rE).$$

This explains the denominator in (1.14). It is obtained by per-location accounting, not by dividing a mass estimate by a distinct count.

5. THE TWO KERNELS AND THEIR CERTIFIED LOCAL INEQUALITIES

5.1. Normalization and the exact energy cost. On $[-1/2, 1/2]$, define

$$(5.1) \quad f_0(s) = \frac{\cos(\sqrt{2}s)}{\sqrt{2}\sin z}, \quad f_1(s) = f_0(s) + u_1 \cos(2\pi s) + u_2 \cos(4\pi s), \quad z = 1/\sqrt{2},$$

and extend both functions by zero outside that interval. The coefficients u_1, u_2 are those in (1.4). Direct integration gives $\int f_0 = \int f_1 = 1$. These are nonnegative densities, not signed test functions: on the interval $\cos(\sqrt{2}s) \geq 3/4$ and $\sqrt{2}\sin z \leq 1$, so $f_0 \geq 3/4$ and $f_1 \geq 3/4 - |u_1| - |u_2| = 71/100 > 0$.

For a real even compactly supported density $f \in L^1 \cap L^2$, define

$$(5.2) \quad \mathcal{R}(f) = \int_{\mathbb{R}} f(s)^2 ds + \iint_{\mathbb{R}^2} |s - t| f(s) f(t) ds dt.$$

The energy of f_0 is R_0 in (1.4). A short calculation also explains why its perturbation cost is exactly quadratic. On the support interval put $W_0(s) = \int |s - t| f_0(t) dt$. Then $W_0'' = 2f_0$ and $f_0'' = -2f_0$. Evenness shows that $f_0 + W_0$ is constant. Its value at $s = 1/2$ is $z \cot z + 1/2 = R_0$, so

$$\mathcal{R}(f_0) = \int f_0(f_0 + W_0) = R_0.$$

If h has integral zero on the interval, the linear variation of $\mathcal{R}(f_0 + h)$ vanishes. Write $H_h(s) = \int_{-1/2}^s h(t) dt$, so H_h vanishes at both endpoints. With $W_h(s) = \int |s - t| h(t) dt$ one has $W_h' = 2H_h$ and hence, by integration by parts,

$$\iint |s - t| h(s) h(t) ds dt = -2 \int H_h(s)^2 ds.$$

For $h(s) = \sum_j u_j \cos(2\pi j s)$, trigonometric orthogonality therefore gives

$$(5.3) \quad \mathcal{R}(f_0 + h) - R_0 = \int h^2 - 2 \int H_h^2 = \sum_j \left(\frac{1}{2} - \frac{1}{4\pi^2 j^2} \right) u_j^2.$$

The functional is quadratic, so there is no omitted higher-order remainder. In particular $\mathcal{R}(f_1) = R_1 = R_0 + \Delta$.

The two pieces of the energy must be kept separately when the support is dilated. Put $A_j = \int f_j^2$ and $B_j = R_j - A_j$ for $j = 0, 1$. Direct integration yields

$$(5.4) \quad A_0 = \frac{1}{4 \sin^2 z} + \frac{z \cot z}{2},$$

$$(5.5) \quad A_1 = A_0 + 2 \sum_{j=1}^2 \frac{(-1)^{j+1} u_j}{2\pi^2 j^2 - 1} + \frac{1}{2} \sum_{j=1}^2 u_j^2.$$

For example, the cross term follows from $\int f_0(s) \cos(2\pi j s) ds = (-1)^{j+1} / (2\pi^2 j^2 - 1)$. If $f_{j,\sigma}(s) = \sigma^{-1} f_j(s/\sigma)$, then

$$(5.6) \quad \mathcal{R}(f_{j,\sigma}) = R_j(\sigma) := \frac{A_j}{\sigma} + B_j \sigma.$$

Both coefficients are positive. The use of A_1, B_1 , rather than the cosine coefficients A_0, B_0 , is necessary for the perturbed short-height calculation.

5.2. The local inputs. Let $k_j = \widehat{f_j}$ with the Fourier convention (2.1). Writing $\text{sinc}(x) = \sin(x)/x$ with value one at zero, their entire formulas are

$$(5.7) \quad k_0(x) = \frac{\text{sinc}(\pi x - z) + \text{sinc}(\pi x + z)}{2\sqrt{2} \sin z},$$

$$(5.8) \quad k_1(x) = k_0(x) + \frac{1}{2} \sum_{j=1}^2 u_j \{ \text{sinc}(\pi(x - j)) + \text{sinc}(\pi(x + j)) \}.$$

Define the seven-point energy for six nonnegative gaps by

$$(5.9) \quad V_{7,j}(g) = \sum_{h=1}^6 \frac{2}{7-h} \sum_{i=1}^{7-h} k_j(g_i + \cdots + g_{i+h-1})^2.$$

The two computer-assisted local inequalities are

$$(5.10) \quad V_{7,0}(g) + \frac{1}{3000} \sum_{i=1}^6 g_i \geq \frac{19}{5000},$$

$$(5.11) \quad V_{7,1}(g) + \frac{1}{500} \sum_{i=1}^6 g_i \geq \frac{7}{400}.$$

Both hold for every $g \in \mathbb{R}_{\geq 0}^6$, including zero gaps and unbounded configurations. Equation (5.10) is Ainta's certificate [2]. Equation (5.11) is the two-mode certificate used for the present union and multiplicity bounds. The supplement gives the rigorous finite verification in detail.

Here is the structure of that verification, to clarify what is asserted. Nonnegative pressure first reduces a possible strict counterexample to a compact simplex. A single-coordinate lower test cuts this down further. Outward scalar enclosures on closed grid cells give lower bounds for every squared-kernel term. A sum of h gaps requires all h final scalar cells; the extra $h - 1$ indices at a summed upper endpoint are included. Adaptive binary subdivision covers the resulting boxes. Interval energy, whole-box Taylor bounds and rigorous positive-definiteness tests each supply sufficient acceptance rules. Removable singularities in (5.7)–(5.8) use convergent power series with explicit remainders. An unaccepted terminal box, a time limit or a failed arithmetic guard gives no certificate. The recorded completed covers exhaust their domains under these rules. This is stronger than checking the objective at grid points, and the proof does not treat a digest or a node count as a substitute for coverage.

In (3.10), these rows give $a = a_j$ and $d = 6p = d_j$, with endpoint debit $B = 6a_j$. The first may be used at $t = 1$. For the second,

$$\frac{a_1}{b_7} = \frac{16807}{16000} > 1, \quad a_1 < 4b_7 < b_7(1 + \sqrt{2(1 - a_1)})^2.$$

Thus its admissible thresholds include the two required in Sections 2 and 4. These are separate uses of its *same* local certificate, not sums of corrections from different densities.

6. FROM THE FINITE OPERATOR TO ZETA ZEROS

6.1. The fixed-test pair-correlation input. We state precisely the analytic theorem being used. If g is a fixed real even integrable function supported in $[-1, 1]$ and Lipschitz at zero, the test-function form of [4, Lemma 5], also stated in [11, Lemma 3.1], is, with $L = \log X$,

$$(6.1) \quad \sum_{0 < \gamma, \gamma' \leq X} \hat{g}\left(\frac{i(\rho - \rho')L}{2\pi}\right) \frac{4}{4 - (\rho - \rho')^2} = \frac{XL}{2\pi} \left\{ g(0) + \int_{\mathbb{R}} |\alpha| g(\alpha) d\alpha + O_g(L^{-1/2}) \right\}.$$

The zeros are counted with multiplicity, and the formula is unconditional. We use it only for fixed smooth functions supported strictly inside $(-1, 1)$. No new proof of (6.1) is claimed here.

Take a real even smooth η supported strictly in $(-\sigma/2, \sigma/2)$, with $0 < \sigma < 1$ and $\int \eta^2 = 1$. Set $f = \eta^2$, $K = \hat{f}$ and $G = f * f$. Both G and G'' are admissible fixed tests. For $w = i(\rho - \rho')L/(2\pi)$, integration by parts gives $\widehat{G''}(w) = -4\pi^2 w^2 K(w)^2$. Consequently the exact identity

$$(6.2) \quad K(w)^2 = \left(\widehat{G}(w) - \frac{\widehat{G''}(w)}{4L^2} \right) \frac{4}{4 - (\rho - \rho')^2}$$

removes the rational weight. Its denominator is nonzero: the real part of the difference of two nontrivial zeros lies in $(-1, 1)$, so that difference cannot be 2 or -2 .

Apply (6.1) *separately* to G and G'' , then combine their sums by (6.2). The second term is $O_f(X/L)$, while the first has leading coefficient

$$G(0) + \int |\alpha| G(\alpha) d\alpha = \mathcal{R}(f).$$

Evenness of f changes the double integral with $|s+t|$ into that with $|s-t|$. Thus, for the multiset

$$(6.3) \quad \mathcal{Z}_X = \left\{ \frac{i(\rho - 1/2) \log X}{2\pi} : 0 < \gamma \leq X \right\},$$

equation (2.3) satisfies

$$(6.4) \quad \frac{H_f(X)}{N(X)} \rightarrow \mathcal{R}(f).$$

The functional equation and conjugation of ζ preserve the multiplicity under $\rho \mapsto 1 - \bar{\rho}$. That map acts as conjugation on (6.3); its real elements are exactly the critical-line zeros. Thus all finite counts above have the interpretations (1.1)–(1.3).

For completeness, the Riemann–von Mangoldt formula gives

$$(6.5) \quad N(X) = \frac{X}{2\pi} \log \frac{X}{2\pi} - \frac{X}{2\pi} + O(\log X) = \frac{X \log X}{2\pi} + O(X),$$

with the usual endpoint convention; see [19] or [4, Lemma 2]. The simple-real span obeys $S \leq X \log X / (2\pi)$, including when it is defined to be zero. Therefore

$$(6.6) \quad \limsup_{X \rightarrow \infty} S/N(X) \leq 1.$$

Only an upper bound is needed, since every span debit is nonnegative.

6.2. Smoothing without changing the test with height. The densities (5.1) have sharp endpoints and cannot simply be inserted as smooth strict-support windows. We now justify their use in the final constants. Fix $j \in \{0, 1\}$, a dilation $0 < \sigma < 1$ and a slack $0 < \delta < 1$. Let $F = f_{j,\sigma}$ and let ψ be a nonnegative even smooth cutoff supported strictly inside $(-\sigma/2, \sigma/2)$. Set

$$(6.7) \quad f' = \frac{\psi^2 F}{\int \psi^2 F}, \quad \eta' = \psi \sqrt{F / \int \psi^2 F}.$$

Choose cutoffs tending to one on every compact interior subinterval, bounded between zero and one. Strict positivity and smoothness of F on that interior make η' smooth after extension by zero. The densities f' tend to F in L^1 and L^2 , and their energies tend to $R_j(\sigma)$; the second part of (5.2) is continuous in L^1 on a common compact support.

If $\tau = \|f' - F\|_1$, their Fourier transforms differ by at most τ on the *real* axis. Both have absolute value at most one there. The sum of all coefficients in a seven-point row is 12, so the squared-kernel energy changes by at most 24τ . Choose the cutoff sufficiently close that $24\tau \leq \delta a_j$. Applying the original local row to the dilated gaps gives

$$(6.8) \quad V_{7,f'}(g) + \sigma \frac{d_j}{6} \sum_{i=1}^6 g_i \geq a'_j := (1 - \delta) a_j.$$

Only real transform arguments occur in a simple-real Gram block, so no uniform comparison at complex zeta arguments is being assumed.

The common-witness transfer now reads

$$(6.9) \quad D_t(M_{f'}) \geq a'_j n - \sigma d_j S - 6a'_j.$$

The pressure has become σd_j ; there is no division by $1 - \delta$. At $j = 0$ use $t = 1$. At $j = 1$ use $t = 2$ for multiplicity two, and for the union recompute

$$(6.10) \quad \Gamma' = 4 + \sqrt{8(1 - a'_1)}, \quad t' = \Gamma'/2 - 1, \quad Z' = \Gamma'^2/8.$$

Here $t' > 2$ and the cap remains valid. The coefficient of u in (2.15) is still exactly zero.

Every application of (6.1) and (6.4) has the smooth density, dilation, slack and threshold fixed before $X \rightarrow \infty$. One then removes the cutoff at fixed slack, sends $\delta \downarrow 0$, and finally sends $\sigma \uparrow 1$. These successive limits of numerical lower bounds require no uniform analytic error as the test changes. An arbitrarily small fixed upper-budget margin for the distinct ratios is handled in the same way below.

7. EVALUATION OF THE GLOBAL PROPORTIONS

7.1. The simple-critical and total-distinct counts. Use the smoothed cosine density in (2.16), with $a = a'_0$, $d = \sigma d_0$ and $B = 6a'_0$. Since $e \geq 0$, the height limit, (6.4) and (6.6) give

$$\liminf_{X \rightarrow \infty} \frac{n(X)}{N(X)} \geq \frac{2 - \mathcal{R}(f') - \sigma d_0}{1 - a'_0}.$$

Take the successive approximation limits described in Section 6.2; this proves (1.8) with its exact constant C .

To obtain the distinct count, substitute the *full* finite inequality into (2.13) before dropping e :

$$\begin{aligned} N^d &\geq \frac{1}{2} \left\{ N + \frac{2N - H - \sigma d_0 S - 6a'_0}{1 - a'_0} + \left(\frac{2}{1 - a'_0} - 1 \right) e \right\} \\ &\geq \frac{1}{2} \left\{ N + \frac{2N - H - \sigma d_0 S - 6a'_0}{1 - a'_0} \right\}. \end{aligned}$$

The coefficient discarded is positive. The same limits now prove (1.9).

7.2. The union and higher-multiplicity mass. Use the smoothed two-mode density and (6.10). Equation (2.15) gives

$$Z'e \leq H + \sigma d_1 S + 6a'_1 - (1 + a'_1)N.$$

For this fixed smooth test it follows that

$$\limsup_{X \rightarrow \infty} \frac{e}{N} \leq \frac{\mathcal{R}(f') + \sigma d_1 - 1 - a'_1}{Z'}.$$

Equation (2.11) and the successive approximation limits give (1.10). Equation (2.12) gives (1.13) for each fixed $r \geq 3$. Although (2.12) is also valid at $r = 2$, the direct estimate (4.5), at $t = 2$ with the same smoothed row, is stronger there and yields (1.11).

7.3. Distinct low-multiplicity locations. A fixed finite budget avoids any uncontrolled division of an $o(N)$ term by N^d . Fix $\xi > 0$ and, for a fixed smoothed two-mode density, set

$$\mathcal{T} = \mathcal{R}(f') + \sigma d_1 + \xi.$$

Equations (6.4) and (6.6), together with $6a'_1/N \rightarrow 0$, show that $H + \sigma d_1 S + 6a'_1 \leq \mathcal{T}N$ for all sufficiently large X . For sufficiently small ξ , slack, cutoff error and $1 - \sigma$, the strict versions of the guards (4.9) hold. Apply the finite ratio (4.10) first, then pass to the height limit and remove ξ and the approximation parameters. This proves (1.12).

For each fixed $r \geq 3$, use $c = \Gamma'/2$ and (4.11)–(4.12). The same procedure and (4.13) give (1.14). The limiting guards can be checked without delicate numerical signs. For example,

$$\frac{5}{4} \leq R_0 < \frac{4}{3}, \quad 0 < \Delta < \frac{109}{250000}, \quad 1 < T_1 < 1.35, \quad 6 < \Gamma < 7$$

give $\beta > 0$, $3 - T_1 - a_1 > 0$, $8 - 2T_1 - a_1 > 0$ and $\alpha = \Gamma - T_1 > 0$. Moreover, using $c^2 = 2Z$ and $\alpha = Z - \beta$,

$$4\alpha + \beta - c^2 = 2Z - 3\beta > 0,$$

since $Z > 5$ and $\beta < 0.35$. The remaining guard $q_c = 0$ is the exact identity (2.17). Thus the strict conditions persist under the small perturbations just used, with this identity retained symbolically. The fine decimal enclosures are independent of these coarse qualitative checks.

One can also see directly why the denominator cannot become too small for a fixed test. Cauchy–Schwarz on the nonzero eigenvalues of A and its rank-one decomposition by distinct locations give

$$(7.1) \quad N^2 \leq \text{rank}(A)H \leq N^d H.$$

Since $H = O_f(N)$, this implies $N^d \gg_f N$. We nevertheless used a fixed upper budget so that the per-location inequalities remain exact throughout the argument.

7.4. Exact arithmetic and strict comparison signs. The numerical constants in the introduction are enclosed by rational intervals, not by floating-point rounding to a presumed number of correct digits. The verifier constructs π from Machin’s identity and alternating arctangent series, bounds square roots by rational squaring, and evaluates sine and cosine with explicit Taylor remainders. Interval division is used only after excluding zero from the denominator. It evaluates (1.4)–(1.7) and the five exact expressions independently of the local cover program. In particular it checks the cap inequalities and the strict margins between each exact lower endpoint and its displayed decimal.

The signs of the improvements over the named basic formulas also have a short analytic proof. Put

$$Z_0 = 3 + 2\sqrt{2}, \quad \eta_* = a_1 - d_1 - \Delta, \quad B = (R_0 - 1)/Z_0.$$

Then $\beta = R_0 - 1 - \eta_*$. Since $a_1 < 1/2$,

$$0 < Z_0 - Z = a_1 \left(1 + \frac{2\sqrt{2}}{1 + \sqrt{1 - a_1}} \right) < 3a_1.$$

Also $\eta_* > 633/125000$, whereas

$$\frac{(R_0 - 1)(Z_0 - Z)}{Z_0} < \frac{a_1}{5} = \frac{7}{2000} < \frac{633}{125000}.$$

It follows that $E < B$, which proves the strict union improvement and the stated higher-cutoff family comparisons. The estimates $1 < R_0 < 4/3$ used here follow from $\cos z \geq 3/4$, $\sin z/z \leq 1$, and, for the upper bound, the alternating-series inequality $\cos z < (5/6)(\sin z/z)$ at $z^2 = 1/2$. For example $\cos z \leq 1 - z^2/2 + z^4/24 = 73/96$, while $(5/6)(\sin z/z) \geq (5/6)(1 - z^2/6) = 55/72 > 73/96$.

The comparison with the mass-two formula $(3 - R_0)/2$ has numerator

$$K_* = a_1(3 - R_0) - 2(\Delta + d_1) > \frac{7}{240} - 2 \left(\frac{109}{250000} + \frac{3}{250} \right) > 0$$

over the positive denominator $2(2 - a_1)$. The comparison with the distinct-location formula $1 - (R_0 - 1)/(8 - 2R_0)$ has numerator $3K_*$ over $(8 - 2R_0)(8 - 2T_1 - a_1) > 0$. These identities explain the gain without relying on a long decimal expansion. The exact scalar calculation supplies the finer margins in Section 1.2.

8. SHORT-HEIGHT INTERVALS WITH FIXED SUPPORT MARGIN

Fix $0 < \theta < 1$ and let $H_X = X^\theta$. In this section all counts are restricted to $X < \gamma \leq X + H_X$, with their definitions and denominators otherwise unchanged. We append a subscript I when needed. For every fixed $0 < \sigma < \theta$ and fixed smooth real even test g supported in $[-\sigma, \sigma]$, Wang’s

Theorem 2.2 [21] states

$$(8.1) \quad \sum_{X < \gamma, \gamma' \leq X + H_X} \hat{g}\left(\frac{i(\rho - \rho') \log X}{2\pi}\right) \frac{4}{4 - (\rho - \rho')^2} \\ = \frac{H_X \log X}{2\pi} \left(g(0) + \int |\alpha| g(\alpha) d\alpha \right) + O_g(H_X + X^\sigma \log^2 X).$$

This is a separate analytic input, not a subtraction of two full-height correlation asymptotics. Its normalized error tends to zero for every fixed strict support margin:

$$\frac{H_X + X^\sigma \log^2 X}{H_X \log X} = O((\log X)^{-1} + X^{\sigma-\theta} \log X) = o(1).$$

Apply (6.2) to the two fixed tests G, G'' as before. The difference form of (6.5) gives

$$N_I(X) = \frac{H_X \log X}{2\pi} + O(H_X + \log X), \quad S \leq \frac{H_X \log X}{2\pi}.$$

It follows that $H_f/N_I \rightarrow \mathcal{R}(f)$ and $\limsup S/N_I \leq 1$. The finite argument is unchanged.

Define

$$(8.2) \quad T_j(\theta) = \frac{A_j}{\theta} + (B_j + d_j)\theta \quad (j = 0, 1), \quad E_\theta = \frac{T_1(\theta) - 1 - a_1}{Z}.$$

Smooth first at a fixed $\sigma < \theta$, take the height limit with every test fixed, remove the cutoff and floor slack, and only then let $\sigma \uparrow \theta$. The resulting lower limiting bounds are

$$(8.3) \quad \liminf \frac{n_I}{N_I} \geq \frac{2 - T_0(\theta)}{1 - a_0}, \quad \liminf \frac{N_I^d}{N_I} \geq \frac{1}{2} \left(1 + \frac{2 - T_0(\theta)}{1 - a_0} \right),$$

$$(8.4) \quad \liminf \frac{U_I}{N_I} \geq 1 - 2E_\theta, \quad \liminf \frac{M_{\leq 2, I}}{N_I} \geq \frac{3 - T_1(\theta)}{2 - a_1},$$

$$(8.5) \quad \liminf \frac{M_{\leq r, I}}{N_I} \geq 1 - \frac{r+1}{r-1} E_\theta \quad (r \geq 3 \text{ fixed}).$$

All lower limits here are as $X \rightarrow \infty$. A negative right side may be replaced by zero. These mass and total-distinct statements hold for each fixed $0 < \theta < 1$ under the stated short-height input, although at small θ some give no nontrivial lower bound.

For distinct low-multiplicity locations, the parameter guards must also survive this substitution. The bound

$$(8.6) \quad \liminf \frac{D_{\leq 2, I}}{N_I^d} \geq 1 - \frac{T_1(\theta) - 1 - a_1}{8 - 2T_1(\theta) - a_1}$$

holds whenever the strict versions of the nonconstant guards in (4.9) hold with $a = a_1$ and $\mathcal{T} = T_1(\theta)$. More explicitly, require

$$T_1(\theta) - 1 - a_1 > 0, \quad 6 - T_1(\theta) > 0, \quad 3 - T_1(\theta) - a_1 > 0, \quad 8 - 2T_1(\theta) - a_1 > 0;$$

the fixed condition $1 - 2a_1 > 0$ already holds. For $r \geq 3$, the analogous formula is

$$(8.7) \quad \liminf \frac{D_{\leq r, I}}{N_I^d} \geq 1 - \frac{E_\theta}{r - 1 - rE_\theta},$$

provided (4.11)–(4.12) hold with $c = \Gamma/2$, $a = a_1$, $\mathcal{T} = T_1(\theta)$, strictly for the nonconstant inequalities and with $q_c = 0$ retained exactly. These conditions hold for θ sufficiently close to one by the global checks and continuity. We do not assert them for every $\theta \in (0, 1)$. The fixed upper-budget construction in Section 7 justifies the distinct denominator here as well. None of these statements is uniform as θ varies with height, and none sets $\sigma = \theta$ inside (8.1).

9. SCOPE OF THE ACCOMPANYING LOCAL EXTENSION

The finite transfer permits other window lengths, profiles and distance-capacity weights, but its premise remains a uniform local inequality. To make that boundary concrete, the supplement also studies a thirteen-point objective with six explicit Fourier modes, two gap pressures and target 31/5000. Scalar exclusion partitions the gap range of a hypothetical strict counterexample into short and long components. Complete local certificates establish the all-long region and every region with exactly one short gap, including all ten interior positions. Their proof uses disjoint triples and an exact convex quadratic lower bound on the remaining small cells.

The resulting lower bounds are 0.006424625 on the all-long region, 0.00623 on the two endpoint-one-short regions and 0.0062 on each interior-one-short region, on the precise domains specified in the supplement. Thus a remaining strict counterexample must have at least two short gaps. Whether such a configuration exists remains unresolved in this work. The full thirteen-point inequality is not asserted, and none of its prospective zero-count consequences is included in (1.8)–(1.14).

The distinction is structural. The common-witness inequality can transfer a proved local floor, and the bounded state correction can transfer a proved subaction. Neither constructs a missing floor or a missing potential. In the present application the two completed seven-point inputs already suffice for every global bound stated in the article. The larger-window result records an additional proved local reduction without enlarging those conclusions.

ACKNOWLEDGMENTS

The author is indebted to his family for their extraordinary patience and support. He is grateful to Dr. C. Oommen and Prof. B. N. Raghunandan of the Department of Aerospace Engineering, Indian Institute of Science, for their mentorship and for fostering the intellectual latitude that enabled this independent research.

NOTE ON AI USE

The author declares the use of consumer-grade Gemini 3.1 and GPT-6 models for prior-art search, developing some ideas and proofs, computer-assisted verification, drafting and LaTeX typesetting; the author takes full and sole responsibility for all contents of this work.

REFERENCES

- [1] L. Alpöge and R. Furman, *More than two thirds of the zeta zeros are simple and on the critical line*, arXiv:2608.13637v2 (2026).
- [2] GitHub contributor **ainta**, *More than 67.3% of the zeros of the Riemann zeta function are simple and lie on the critical line*, public research repository, commit 040c5e8 (2026).
- [3] F. Aryan, *On an extension of the Landau–Gonek formula*, J. Number Theory **233** (2022), 389–404, doi:10.1016/j.jnt.2021.06.015.
- [4] S. A. C. Baluyot, D. A. Goldston, A. I. Suriajaya and C. L. Turnage-Butterbaugh, *An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta-function*, Acta Arith. **214** (2024), 357–376, doi:10.4064/aa230612-20-3.
- [5] S. A. C. Baluyot, D. A. Goldston, A. I. Suriajaya and C. L. Turnage-Butterbaugh, *Pair Correlation of Zeros of the Riemann Zeta Function I: Proportions of Simple Zeros and Critical Zeros*, arXiv:2501.14545v3 (2026).
- [6] E. Carneiro, V. Chandee, F. Littmann and M. B. Milinovich, *Hilbert spaces and the pair correlation of zeros of the Riemann zeta-function*, J. Reine Angew. Math. **725** (2017), 143–182, doi:10.1515/crelle-2014-0078.
- [7] A. Chirre, F. Gonçalves and D. de Laat, *Pair correlation estimates for the zeros of the zeta function via semidefinite programming*, Adv. Math. **361** (2020), article 106926, doi:10.1016/j.aim.2019.106926.
- [8] M. Devine, *An Unconditional 67.3399% Bound and Conditional Advances Beyond 67.92% for Simple Critical Zeros of the Riemann Zeta Function*, Zenodo preprint, version 1.0.3 (2026), doi:10.5281/zenodo.22066689.
- [9] F. Gonçalves, D. de Laat and N. Leijenhorst, *Multiplicity of nontrivial zeros of primitive L-functions via higher-level correlations*, Math. Comp. **94** (2025), no. 354, 2041–2058, doi:10.1090/mcom/4005.
- [10] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), no. 8, 1281–1292, doi:10.1109/TC.2017.2690633.

- [11] Y. Lamzouri, *A new proof that more than 2/3 of the zeros of the Riemann zeta function are simple and on the critical line*, arXiv:2609.02882v2 (2026).
- [12] T. Lince / Zeta Lab, *Results: where the seven-point certificate ends*, public research report, commit 2da62eb (2026).
- [13] GitHub contributor `mdumitrean`, public union extension, formal-math pull request 17, accessed 10 September 2026.
- [14] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in Analytic Number Theory (St. Louis, 1972), Proc. Sympos. Pure Math. **24**, Amer. Math. Soc., 1973, 181–193, doi:10.1090/pspum/024/9944.
- [15] H. L. Montgomery, *Distribution of the zeros of the Riemann zeta function*, in Proceedings of the International Congress of Mathematicians (Vancouver, 1974), Vol. 1, Canad. Math. Congress, 1975, 379–381.
- [16] V. Ojha, *A certified 67.32001% lower-bound candidate for simple zeros of the Riemann zeta function on the critical line*, public research draft, commit 1610b97 (2026).
- [17] Y. Shi, *A two-certificate trace–energy deduction for simple zeros of the Riemann zeta function*, public research draft, version 0.1.0, commit 1469eee, doi:10.5281/zenodo.21926962 (2026).
- [18] A. Simonič, T. S. Trudgian and C. L. Turnage-Butterbaugh, *Some explicit and unconditional results on gaps between zeroes of the Riemann zeta-function*, Trans. Amer. Math. Soc. **375** (2022), no. 5, 3239–3265, doi:10.1090/tran/8571.
- [19] E. C. Titchmarsh, *The theory of the Riemann zeta-function*, second edition, revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986.
- [20] Z. Waddle, *Unconditional low-multiplicity profiles for zeros of the Riemann zeta function*, public unrefereed research draft, operative theorem statement, accessed 10 September 2026.
- [21] B. Wang, *Simple critical zeros and distinct zeros of the Riemann zeta-function in short intervals*, arXiv:2609.07918v1 (2026).

DEPARTMENT OF AEROSPACE ENGINEERING, INDIAN INSTITUTE OF SCIENCE

Email address: `arun.chandru@gmail.com`